

Multiple Choice

1. **Answer:** E

Options: A) Humans cannot convert any fatty acids into docosahexaenoic acid; B) Flaxseeds are a rich source of docosahexaenoic acid; C) Docosahexaenoic acid levels are consistently higher in vegans than in omnivores; D) Vegan diets provide an abundance of docosahexaenoic acid naturally; E) Lack of docosahexaenoic acid in vegans causes visual and cognitive impairment

Note: The statement is not true because while docosahexaenoic acid (DHA) is important for visual and cognitive function, vegans can obtain sufficient DHA through algal supplements or by converting alpha-linolenic acid (ALA) from plant sources into DHA, mitigating the risk of impairment.

2. **Answer:** A

Options: A) 96 hours.; B) 36 hours.; C) 60 hours.; D) 2 weeks.; E) 1 week.

Note: The correct answer is 96 hours because current clinical guidelines recommend that peripheral cannulas should ideally be replaced every 72 to 96 hours to reduce the risk of infection and maintain optimal venous access.

3. **Answer:** C

Options: A) apoA, apo(a), apoB, apoC and apoH; B) apoA, apoB, apoC, apo E, and apoL; C) apoA, apo(a), apoB, apoC and apoE; D) apoA, apoB, apoC, apoD, and apoF; E) apoA, apoB, apoC, apo(a) and apoM

Note: The correct answer identifies the five main series of apoproteins-apoA, apo(a), apoB, apoC, and apoE-because they play crucial roles in lipid metabolism and transport, serving as structural components of lipoproteins and influencing cholesterol levels and cardiovascular health.

4. **Answer:** C

Options: A) Aunt; B) Nephew; C) Brother; D) Grandchild; E) Great-grandchild

Note: The recurrence risk for schizophrenia is greatest for a brother of an affected individual because siblings share approximately 50% of their genetic material, making them more susceptible to inheriting the genetic risk factors associated with the disorder compared to more distantly related relatives.

5. **Answer:** A

Options: A) the muscle is fully hydrated.; B) oxygen levels in the muscle are low.; C) glucose levels in the muscle are low.; D) oxygen is readily available.; E) the pH of the muscle falls.

Note: Muscle lactate production increases when the muscle is fully hydrated because adequate hydration supports optimal enzymatic activity and energy metabolism during intense exercise, facilitating the anaerobic glycolytic pathway that produces lactate.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Biotin acts as a coenzyme for carboxylase enzymes, specifically pyruvate carboxylase, which catalyzes the conversion of pyruvate to oxaloacetate in gluconeogenesis and the citric acid cycle.

7. **Answer:** True

Options: A) True; B) False

Note: The total daily dose of Cotrimoxazole at 960 mg every 12 hours is 1.92 grams because administering 960 mg twice a day results in a total of 1920 mg, which is equivalent to 1.92 grams.

8. **Answer:** False

Options: A) True; B) False

Note: The most reliable method to assess for clubbing involves examining the angle between the nail and the nail bed from above, as this provides a clearer indication of changes associated with clubbing compared to inspecting from below.

9. **Answer:** False

Options: A) True; B) False

Note: Grandparents generally value meaningful, personal interactions and quality time with their grandchildren over the frequency of text messages, making the statement false.

10. **Answer:** False

Options: A) True; B) False

Note: Dental plaque is more likely to accumulate on the buccal surfaces of the upper molars and the lingual surfaces of the lower incisors due to their anatomical positions and the way food particles and saliva flow, rather than on the palatal and labial surfaces.

Long Form Response

11. **Answer:** The relationship between optimism and longevity in older adults is significant, as research suggests that optimistic individuals tend to experience better health outcomes and longer lifespans compared to their pessimistic counterparts. Pessimism can adversely affect health by increasing stress levels, promoting unhealthy behaviors, and diminishing resilience in the face of adversity, all of which can lead to a higher risk of chronic illnesses and premature death. Studies have shown that many pessimists die at a younger age, highlighting the detrimental impact of a negative outlook on life. In contrast, optimism is associated with lower rates of depression, enhanced coping strategies, and greater social support, contributing to improved overall well-being and longevity. Therefore, fostering an optimistic mindset in older adults may be crucial for promoting health and extending lifespan.

Note: correct because it highlights the positive correlation between optimism and health outcomes in older adults, while also detailing how pessimism negatively impacts physical and mental health, ultimately affecting longevity.

12. **Answer:** Amino acids play a crucial role in gluconeogenesis, serving as substrates that can be converted into glucose. However, tryptophan cannot serve as a substrate in this metabolic pathway due to its unique catabolic fate. Unlike other amino acids that can be directly transformed into intermediates such as pyruvate or oxaloacetate for gluconeogenesis, tryptophan is primarily metabolized through the kynurenine pathway, leading to the production of metabolites that do not enter the gluconeogenic route. Additionally, tryptophan's degradation results in the formation of acetyl-CoA and other non- gluconeogenic compounds, which limits its utility in glucose synthesis. Therefore, the specific metabolic pathway of tryptophan highlights its inability to directly contribute to gluconeogenesis.

Note: correct because tryptophan is uniquely metabolized through the kynurenine pathway, resulting in the production of metabolites that do not align with the intermediates required for gluconeogenesis, unlike other amino acids that can be converted into gluconeogenic substrates.